12 Design Your Own Algorithm

AGES 8-12

MASCOT GOES HERE

KID INSTRUCTIONS

You've made it to the **FINAL activity**! You've learned sequences, loops, conditionals, sorting, debugging, and patterns. Now it's your turn — **YOU'RE the algorithm designer**. Pick a real problem and solve it your way!

Design

Design — To plan and create something from scratch. Real programmers design algorithms — the same way architects design buildings.
Example: Every app on your phone started as a design — someone thought about what it should do BEFORE they built it!

The 5-Step Algorithm Design Process

Real programmers follow 5 steps every time they design an algorithm. Follow these steps yourself throughout this activity!

1. **PICK YOUR PROBLEM** — What task do you want to solve?
2. **LIST THE STEPS** — Break it down into small actions
3. **LOOK FOR PATTERNS** — Can you use a LOOP? A CONDITIONAL?
4. **WRITE THE ALGORITHM** — Combine steps + patterns
5. **TEST AND FIX** — Try it out. Find bugs. Improve.

Fun fact: These same 5 steps are used by game designers, AI researchers, and the engineers at Google. When YOU use them, you're thinking like they do.

Basic Version

Your First Algorithm Design — Pick ONE task from below (or invent your own!). Then follow the 5 steps to design your algorithm.

1. Making a Bowl of Cereal
Walk through every step from empty bowl to eating.
Hint: Watch the ORDER — like Activity 2!

2. Getting Dressed for School
From pajamas to fully dressed and ready.
Hint: What comes FIRST — socks or shoes? SEQUENCE matters!

3. Feeding 3 Fish in a Tank
Give each fish exactly one pinch of food.
Hint: This has repetition — try using a LOOP!

STEP 1 – My chosen task is:

STEP 2 – List the steps
(break it down into small actions)

1.
2.
3.
4.
5.

STEP 3 – Look for patterns
Do any steps repeat? Are there Yes/No questions? Circle the tools you'll use:

SEQUENCE LOOP CONDITIONAL NESTED LOOP

STEP 4 – Write your final algorithm
(combine steps + patterns)

STEP 5 – Test it!
Ask someone to follow your algorithm exactly. What happened? Any bugs?

Challenge Version

Design an Algorithm from Scratch — This time, YOU pick the problem. Choose something you do regularly — a chore, a game routine, a homework habit. Design a full algorithm for it, using at least **ONE loop** AND at least **ONE conditional**.

My problem is:

Why I chose this problem:

My algorithm
(write in pseudocode — use IF, REPEAT, and shorthand you learned)

Which tools did I use?
(circle all that apply)

○ Sequence ○ Loop ○ Nested Loop ○ Conditional ○ Nested Conditional ○ Pattern Recognition

Test it!

I tested my algorithm by:

What bug did I find (if any)?

How did I fix it?

Self-Check

- ○ I picked a real problem and designed an algorithm for it
- ○ I used at least ONE tool from earlier activities (loop, conditional, etc.)
- ○ I tested my algorithm and improved it if needed
- ○ I can explain my algorithm to someone else in my own words
- ○ I feel like an algorithm designer now!

Reflection Box

Look back at every activity you did in this workbook. Which concept was your FAVORITE? Which was the HARDEST? What surprised you the most about how algorithms work?

Bonus: Teach Someone Else

The best way to know you understand something is to TEACH it. Find a family member, friend, or even a pet — and teach them ONE concept from this workbook. Explain it in your own words.

The concept I taught: (circle one)

○ Algorithm ○ Sequence ○ Loops ○ Nested Loops ○ Conditionals ○ Nested Conditionals ○ Debugging ○ Bubble Sort ○ Selection Sort ○ Pattern Recognition ○ Grid Navigation

Who I taught:

How I explained it (in my OWN words):

Did they understand? What questions did they ask?

★ CONGRATULATIONS! ★

You've completed the workbook.

You're officially an ALGORITHM THINKER!

TEACHER / PARENT NOTE

This capstone activity is where kids stop being students following exercises and become designers creating from scratch. The 5-step design process they used (pick problem, list steps, look for patterns, write, test) is the same process used by real software engineers — from Google engineers designing search algorithms to game developers designing enemy AI. This workbook has given your child the mental toolkit of computational thinking, a skill that transfers directly to math, science, writing, and everyday problem-solving. Discussion prompt: Ask your child what problem they'd love to solve next — maybe a chore, a game, a homework habit. Encourage them to KEEP DESIGNING ALGORITHMS beyond this workbook. This is just the beginning.